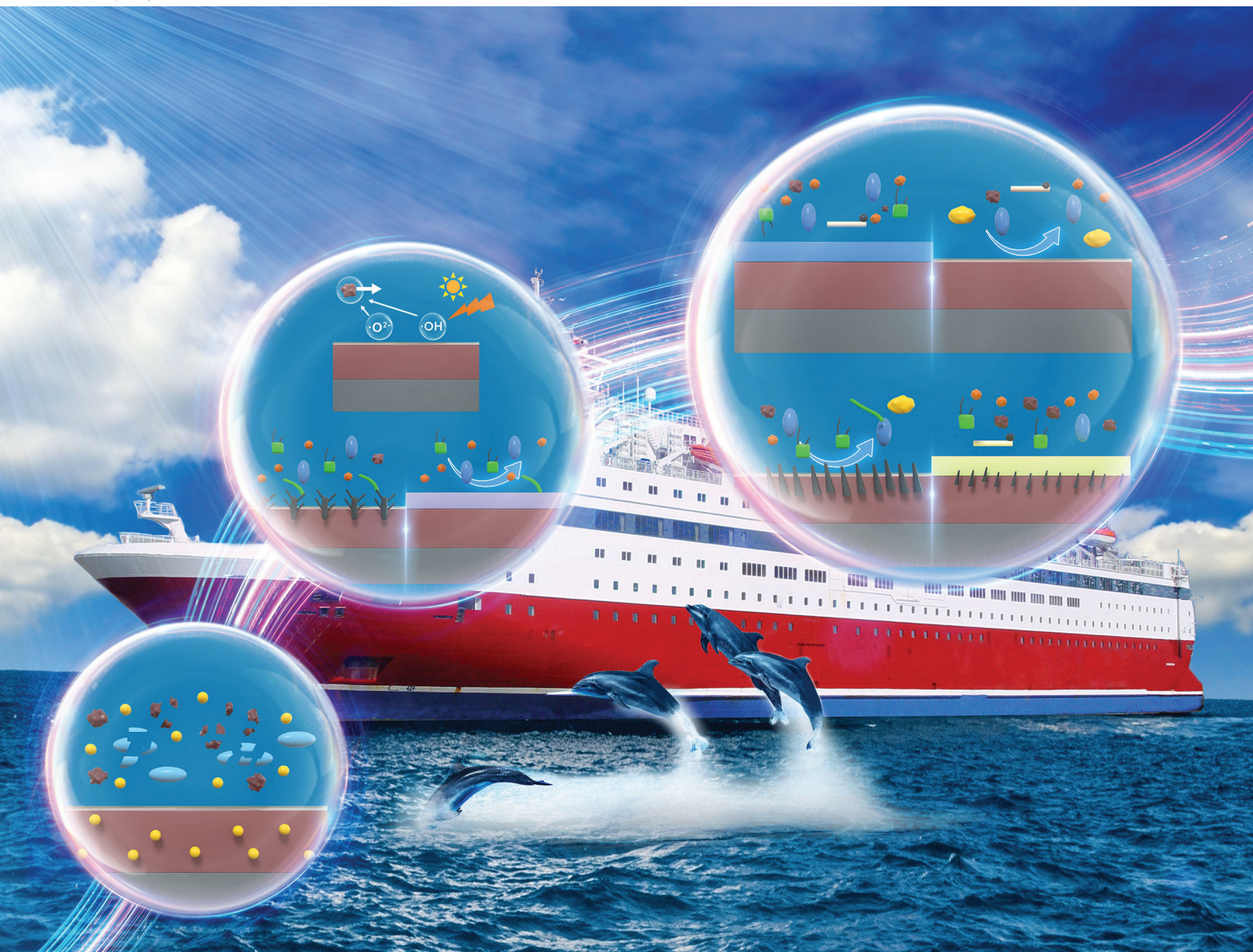


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Xing Zhou *et al.*

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Research progress of environmentally friendly marine antifouling coatings

Mengyue Liu, , Shaonan Li, Hao Wang, Rijia Jiang and Xing Zhou *

Marine biofouling has brought a serious impact on human marine production and transportation activities. Applying antifouling coatings is an effective method to prevent marine biofouling. Here, we have conducted a systematic review of the development of environmentally friendly marine antifouling coatings in the past three years. Firstly, we briefly introduce **marine fouling organisms and their growing process** and then classify **antifouling coatings into 5 parts according to different antifouling mechanisms**. While analyzing the advantages of the coating, we also point out the existing problems in the research of the material. Finally, we present the directions of antifouling coatings in the future, aiming to provide new ideas for developing new antifouling coatings and technologies.

1. Introduction

The ocean contains immeasurable resources; therefore the exploration of the ocean has become the strategic development goal of many countries. Thus it is of great significance for global economy and social civilization to develop marine science and technology.

Ships, submarines, offshore platforms, and offshore drilling facilities are important parts of maritime transportation and ocean exploration. However, a large number of marine organisms are often attached to the surfaces of the equipment when they are in contact with seawater for a long time. Such attachment will have a serious impact on the equip-

ment, especially in the shipping industry. **The harmful attachment will increase the surface roughness of the hull and fuel consumption, accelerate corrosion of underwater facilities, and shorten service life.**^{1,2} In addition, the fouling organisms attached to the surface of the ship would be taken into another area as the ship moves, **causing the invasion of alien species and destroying the local ecological balance.** This kind of undesirable deposition caused by the attachment of marine organisms is called marine biofouling.³ So far, more than 5000 marine fouling organisms have been found, including bacteria, diatoms, barnacles, bryozoans, mussels and algae.⁴ According to incomplete statistics, the cost to the world's annual economic losses caused by marine biofouling is more than hundreds of billions of dollars.⁵ Therefore, marine pollution prevention is of great significance to exploration and utilization of marine resources.⁶

School of Chemistry and Life Sciences, Suzhou University of Science and Technology, Suzhou 215009, China. E-mail: xzhou@mail.usts.edu.cn; Fax: +86-512-68056461; Tel: +86-512-68056461



Mengyue Liu

Mengyue Liu comes from Suqian, China. She received a bachelor's degree in Applied Chemistry from Xuzhou University of Technology in 2019. Then, she started her master's course under the supervision of Prof. Zhou at Suzhou University of Science and Technology. Her research work focuses on the synthesis of antifouling coatings.



Shaonan Li

Shaonan Li comes from Yangzhou, China. He received a bachelor's degree in Applied Chemistry from Nanjing Xiaozhuang University in 2018. Then, he started his master's course under the supervision of Prof. Zhou at Suzhou University of Science and Technology. His research work focuses on the synthesis of self-healing polymers and their sensing applications.

There are mainly four marine antifouling methods including the mechanical removal, electrochemical method, ultrasonic method and using antifouling coatings. The mechanical removal method is to regularly use tools to scrub away the fouling organisms. The electrochemical method is to electrolyze seawater to produce hypochlorous acid or metal oxide with antifouling activity. The ultrasound method is the use of high-frequency sound waves to kill microorganisms. The application range of these three antifouling methods is small owing to the high cost and poor efficiency. Therefore, the method of applying antifouling coatings stands out as a result of the simple operation, low cost and high efficiency.⁷ In the 1970s, tributyltin self-polishing coating (TBT-SPC) was the most successful marine antifouling coating.^{8,9} However, in the 1980s, studies found that TBT-SPC is toxic, and with its gradual release, the accumulation of tin in seawater will cause genetic variation and deformity of marine organisms and may even enter the human body through the seafood chain, endangering people's health.¹⁰ Thus, the International Maritime Organization (IMO) banned the use of TBT as an antifoulant for marine coatings on January 1, 2008.³ Therefore, researchers paid more attention to the development of environmentally friendly marine antifouling coatings.

This article classifies environmentally friendly marine antifouling coatings in the past 3 years according to the different antifouling mechanisms of the coatings. The focus of it is on the development status and the antifouling mechanism of chemical antifouling coatings, physical antifouling coatings and novel antifouling coatings. On this basis, it also evaluates the advantages and disadvantages of the materials. Finally, the future direction of marine antifouling coatings is prospected.

2. Formation mechanism of biofouling

Biofouling is a common spontaneous biological hazard in the shipping and marine economic industries. According to the most widely accepted theory, the formation of most marine biofouling mainly includes 4 stages (Fig. 1).¹¹

2.1. Conditioned film

When a solid substrate is immersed in water for 1 minute, organic molecules such as polysaccharides, proteins, glycoproteins and some inorganic compounds in the seawater are deposited on the surface of the substrate through interaction forces such as van der Waals forces, hydrogen bonds, and static electricity to form the conditioned film.¹²

2.2. Biofilm

Bacteria, diatoms and other microorganisms gather on the conditioned film and then bond with the conditioned film through extracellular macromolecules to form a biofilm.

2.3. Attachment of small fouling organisms

Because a biofilm provides good living conditions for small multicellular fouling organisms such as diatom spores, the surface is covered with a mucus layer within a few days.^{13,14}

2.4. Attachment of large fouling organisms

Finally, a wide variety of large fouling organisms such as barnacles, mussels, bryozoans, coilworms, and macroalgae will attach and grow on the surface.¹⁵

The above process can be completed within one or two months and can cover the surface of the equipment for several years. However, due to the diversity of the marine environment and complex habits of marine organisms, not all the attachments of marine fouling organisms go through all the processes in the above theory. For example, the spores of the large green alga *Ulva* and the larvae of barnacles can be attached to the non-fouling surface without the presence of a biofilm and mucus in sea water.^{16,17}

3. Environmentally friendly marine antifouling coating

Marine antifouling is the use of physical or chemical methods to prevent marine fouling organisms from attaching and growing on the surface of the equipment, or making them easy to detach from the surface. At present, there are mainly



Hao Wang

Hao Wang comes from Jiangsu, China. He received a bachelor's degree in Applied Chemistry from Taizhou University in 2019. Then, he started his master's course under the supervision of Prof. Zhou at Suzhou University of Science and Technology. His research work focuses on the synthesis of electromagnetic interference shielding materials and their applications.



Rijia Jiang

Rijia Jiang comes from Hunan, China. He received a bachelor's degree in Applied Chemistry from Suzhou University of Science and Technology in 2019. Then, he started his master's course under the supervision of Prof. Zhou at Suzhou University of Science and Technology. His research work focuses on the synthesis of self-healing polymers and their applications.

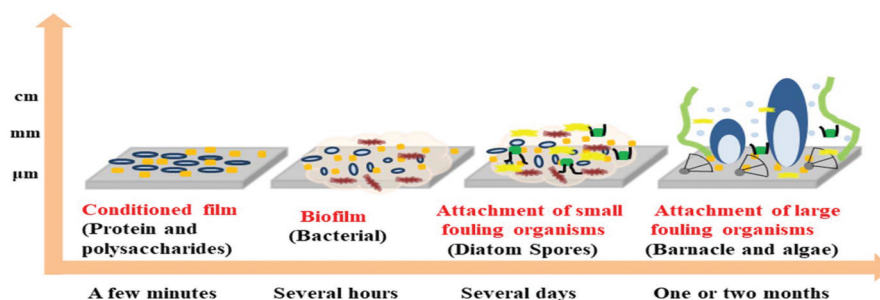


Fig. 1 The growing process of marine biofouling and main fouling organisms.

the following 3 categories including 5 types of marine antifouling coatings (Table 1).

4. Chemical antifouling

4.1. "Chemical killing" antifouling coating

"Chemical killing" antifouling coatings mainly refer to the coatings added with antifouling agents. These coatings are simple to construct and cost-effective and thus currently occupy more than 90% of the global antifouling coating market. These coatings are mainly composed of polymer resin, an antifouling agent, a solvent, a filler and additional auxiliary materials. Organotin antifouling coatings (TBT) were widely used in marine antifouling fields due to their widespread and highly effective antifouling effects in the early days. However, the use of TBT will cause irreversible harm to the marine environment.^{18,19} Therefore, the use of TBT was completely banned in 2008.²⁰ Since then, some environmentally friendly antifouling agents such as copper-based compounds and natural product antifouling agents have been favored by researchers.

4.1.1. Copper-based compound as the antifoulant. Organic copper materials are considered as an ideal substitute for organotin

owing to their low cost, remarkable antifouling performance and low toxicity (1/40–1/50 of organotin).²¹ At present, the widely used antifouling coatings are composed of Cu_2O as an antifoulant. The copper ions produced by Cu_2O in seawater can inactivate the main enzymes that sea creatures rely on for survival, causing biological tissues to change and die, thus possessing antifouling properties.³ In practical applications, it can be found that the release rate of Cu^{2+} is very fast at the initial stage. In order to ensure the service life of the coating, the actual amount of organic copper antifoulant added to the coating is often much higher than the theoretical value. There have been phenomena of excessive Cu^{2+} content and marine pollution in sea areas where copper-based antifouling coatings are intensively used, such as ports.²² Therefore, in recent years, the main direction of research studies has been to reduce the amount of organic copper antifoulants and slow down the Cu^{2+} release rate without affecting the antifouling effect of the coatings.^{23–25} For example, Mao *et al.*²⁶ prepared an antifouling coating with embedding Cu_2O into polyvinylpyrrolidone microcapsules (PVP- Cu_2O) as the antifouling agent (Fig. 2a) and hydrophilic polyurethane/epoxy resin (PU/EP) as the matrix resin. Compared with the cuprous oxide antifouling coating, the PVP- Cu_2O antifouling coating can improve the dispersion of the coating (Fig. 2b) and effectively reduce the release rate of the antifouling agent (Fig. 2c), thereby extending the service life of the coating.

Although organic copper antifouling coatings have been widely used in marine equipment, there are still potential environmental risks.²⁷ The long-term accumulation of the copper element will cause the death of a large number of algae and destroy the marine ecological balance since the copper element of Cu_2O will be released into the marine environment; thus this type of coating will eventually be restricted or prohibited.^{28,29} Therefore, the development of non-toxic, efficient, and environmentally friendly new chemical antifouling coatings is the trend to prevent marine biological fouling.

4.1.2. Natural product antifouling agent. In order to prevent the addition of chemical antifouling agents from damaging the ecological environment, researchers are trying to find useful substances from nature to replace these chemical antifoulants. Many organisms in nature contain active substances themselves, which have natural chemical defense



Xing Zhou

Xing Zhou received his bachelor's degree in polymer materials and engineering from the North University of China in 2006 and his doctor's degree in Applied Chemistry from the North University of China in 2012. Then, he carried on his research career at Suzhou University of Science and Technology and was promoted to Professor in 2017. His research interests are focused on inorganic nano-

materials, intelligent self-healing polymer materials and flexible devices, and energy and environmental materials.

Table 1 Types of marine antifouling coatings

Type	Details	Mechanism	Advantages	Disadvantages	Ref.
Chemical antifouling "Chemical killing" antifouling coating	1. Copper-based compound	The release of copper ions kills biological tissues to inhibit their growth	Inexpensive and widely used	Accumulation of copper ions harms the marine environment	18–29
	2. Natural product	Inhibit the attachment and growth of marine organisms	Non-toxic and environmentally friendly	High cost and complicated extraction process	30–66
	3. Chemically synthesized organic antifoulant	Destroy biological cells and kill fouling organisms	Non-pollution and large-scale production	Severe antifouling agent dialysis and short antifouling effect	67–73
Physical antifouling Fouling release antifouling coating	1. Silicone	Low surface energy makes pollution easy to desorb from the surface	Wide application and price advantage	Poor mechanical properties and poor antifouling performance under static conditions	74–81
	2. Organic fluorine	A hydration layer is formed on the surface to reduce the adhesion of fouling organisms	Good chemical stability	Poor antifouling performance in practical applications	82–98
	1. Polyethylene glycol polymer 2. Hydrogel polymer 3. Zwitterionic polymer	Weak adhesion of foulings	Non-toxic	High cost and short period of validity	99–110
Bionic coating	1. Micro-nano structured surface	Super slippery surface make it difficult for pollutants to attach	Good chemical stability	Short antifouling effective period	111–120
	2. Slippery liquid-infused porous surfaces	Hydrophobicity and sterilization ability	Good mechanical properties	Material compatibility needs to be improved	121–131
	3. Amphiphilic antifouling coating	Free radicals can degrade pollutants and bacteria	Good cycling stability	Low visible light utilization rate	132–136
Novel antifouling materials Novel composite coating	1. Nanocomposite antifouling coating	Hydrophobicity and sterilization ability	Good antifouling performance	Anti-fouling mechanism needs in-depth study	137–143
	2. Photocatalytic antifouling coating	Hydrophilicity and hydrophobicity reduce the adhesion of organisms			

mechanisms against biological pollution.^{30–33} Natural product antifouling coatings are prepared by extracting natural products with antifouling activity from these organisms, which have always been regarded as the most environmentally friendly and promising antifouling coatings.

4.1.2.1. *Marine products with antifouling properties.* Many natural substances such as indole compounds, chitosan and lactones can be produced from the marine creatures and bacteria. These compounds have certain marine antifouling effects and are harmless to the marine environment.^{34–40}

For example, there are a lot of indole derivatives in bryozoans and ascidians. In order to explore the antifouling properties of indole derivatives, Feng *et al.*⁴¹ synthesized 7 kinds of indole derivatives, and the results of the antifouling test proved that indole derivatives have excellent antifouling properties. The experiment also found that the antifouling effect of coatings with indole derivatives as the antifoulant is poor because the release rate of indole derivatives in coatings is uncontrollable. Therefore, Feng *et al.*⁴² did an in-depth study and came up with the introduction of indole derivatives into macromolecules, which can ensure long-term antifouling effects. Subsequently, Ni *et al.*⁴³ introduced the indole derivative into the acrylic acid metal salt resin to prepare an antifouling coating, and the antifouling mechanism of the coating was explored (Fig. 3). First of all, indole derivatives can interfere with the balance of Ca^{2+} in algae cells and reduce the abundance of intracellular Ca^{2+} .^{44,45} Ca^{2+} is an essential ion in almost all aspects of the actual algae physiology during the growth and development of algae, so indole derivatives will inhibit the growth of algae.^{46,47} In addition, the self-polishing properties of the acrylic metal salt resin can continuously update the coating surface through metal ion exchange and the dissolution of hydrophilic groups. In this way, the fouling organisms attached to the surface of the coating will fall off, and new indole derivatives will appear on the new surface of the coating. Therefore, the synthetic resin combines the high biological activity of indole derivatives and the self-polishing performance of acrylic metal salt resin and has better antifouling performance especially in inhibiting algae. This experiment conducted an antifouling test on 4 kinds of algae. The data in Fig. 4 describe that the more indole derivatives introduced in the resins, the higher the corresponding inhibition rates are. The results of the marine antifouling test (Fig. 5) are consistent with the results of algae inhibition rates.

Chitosan is one of the most abundant biopolymers on the earth. It is a completely biodegradable and biocompatible natural polymer that can be used as an adhesive, antibacterial agent and antifungal agent.^{48–50} Studies have found that a single coating formed by only adding chitosan has defects in mechanical strength and barrier properties, while the molecular chain of chitosan contains functional groups such as amino and hydroxyl groups that can support substances with different properties to prepare functional materials.^{51,52} For instance, a composite film can be formed by the combination of chitosan and nano-metal materials, which not only has the

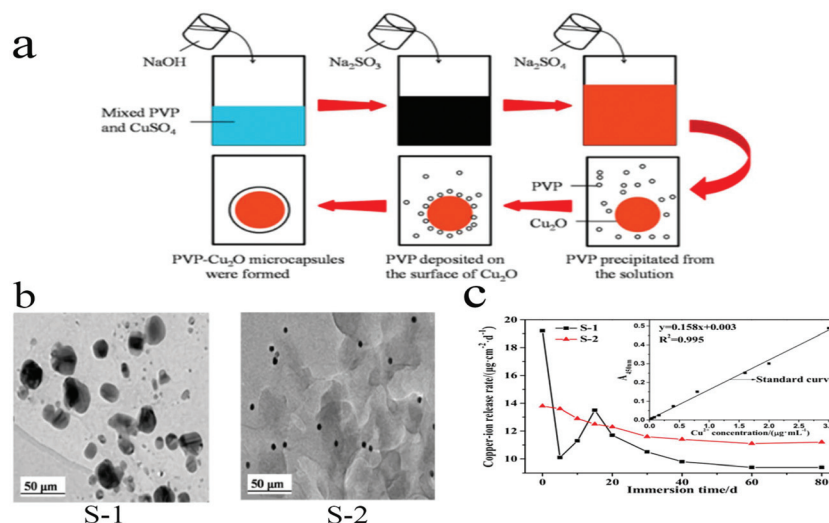


Fig. 2 (a) Schematic diagram of the synthesis process of PVP-Cu₂O microcapsules, (b) TEM images of sample S-1 (ordinary Cu₂O antifouling coating) and sample S-2 (modified Cu₂O antifouling coating), and (c) standard curve and copper-ion release rate curves of samples S-1 and S-2 (reproduced with permission from ref. 26, Copyright 2020, Elsevier).

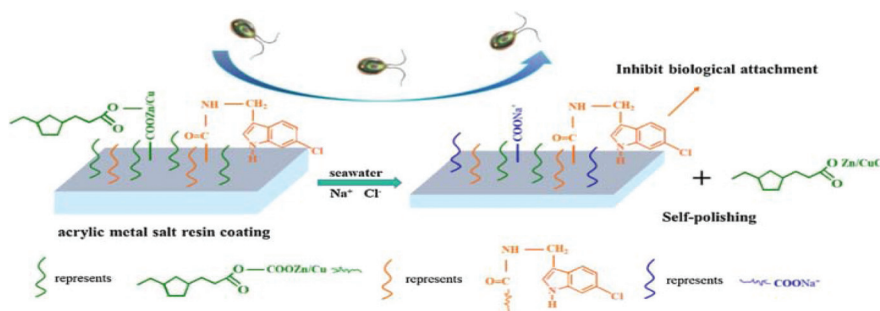


Fig. 3 Antifouling mechanism diagram of acrylic metal salt resins (reproduced with permission from ref. 43, Copyright 2020, Elsevier).

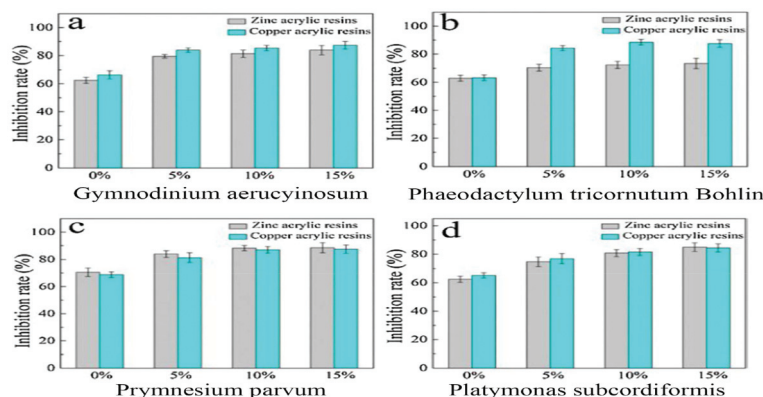


Fig. 4 The algal inhibition rates of resins: *Gymnodinium aerucyinosum* (a), *Phaeodactylum tricornutum* Bohlin (b), *Prymnesium parvum* (c), and *Platymonas subcordiformis* (d). (Reproduced with permission from ref. 43, Copyright 2020, Elsevier).

flexibility of organic matter, but also the rigidity and stability of inorganic matter.^{53,54} Thus, El-saied *et al.*⁵⁵ extracted chitosan from shrimp shells and synthesized nano-chitosan coated ZnO nanoparticles (NCCZO NPs), and then NCCZO NPs were

incorporated into commercial coatings for marine hanging board experiments. The results of antifouling experiments show that NCCZO NPs have 90% antifouling effect on marine bacteria, and coatings with NCCZO NPs have better bioimpe-

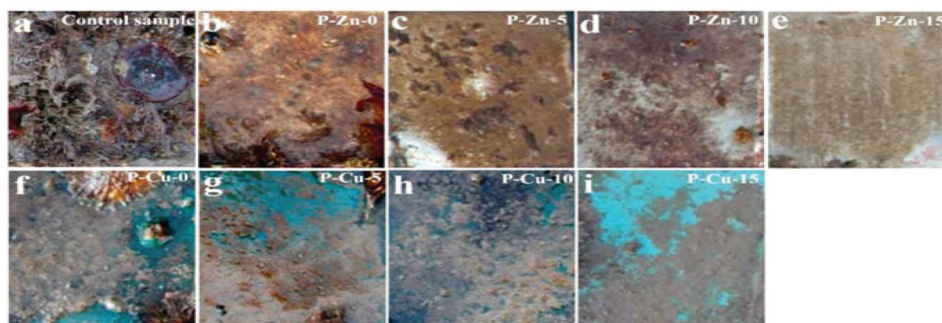


Fig. 5 Images of coatings after immersion in the sea for six months, (a) control sample, (b) P-Zn-0, (c) P-Zn-5, (d) P-Zn-10, (e) P-Zn-15, (f) P-Cu-0, (g) P-Cu-5, (h) P-Cu-10, (i) P-Cu-15. The number represents the mole fraction of the indole derivative introduced. (Reproduced with permission from ref. 43, Copyright 2020, Elsevier).

dance effects. Therefore, chitosan is one of the ideal choices for new antifouling agents.

Butene lactone extracted from marine bacteria is a promising natural antifouling agent with high antifouling activity, low toxicity and fast degradation. More importantly, it has a simple structure and can be synthesized on a large scale with low cost through chemical methods.⁵⁶ In order to better control the release rate of the antifouling agent and prolong the service life of the coating, Pan *et al.*⁵⁷ added butene lactone to polylactic acid-based polyurethane to synthesize an environmentally friendly coating. The antifouling results described that **butene lactone in the coating can be released continuously for 3 months, and coating with 84% polymer and 16% butene lactone has remarkable antifouling ability with only 7% area of panel covered by fouling.**

At present, marine biological antifouling agents cannot be commercially produced as a result of cumbersome extraction steps, high costs, and inability to obtain sufficient amounts from the ocean. However, we can explore their antifouling mechanism through chemical methods, which is of great significance for the exploitation of new environmentally friendly antifouling agents.

4.1.2.2. Terrestrial products with antifouling properties. Terrestrial organisms are also valuable sources of natural anti-

fouling agents. Compared with marine organisms, many terrestrial plants are easy to commercialize owing to their widespread distribution or large-scale cultivation.⁵⁸

Capsaicin is a biologically active substance extracted from pepper, which can inhibit the attachment and growth of attached organisms. It is currently recognized as a natural non-toxic antifouling agent.^{59–62} For example, Wang *et al.*⁶³ conducted a series of studies on capsaicin and its derivatives and synthesized 6 kinds of capsaicin derivatives for the first time. The results of the antifouling test verify the excellent antifouling performance of capsaicin and its derivatives. Then, Hao *et al.*⁶⁴ combined capsaicin (CAP) and chitosan (CS) to form nanocapsules and introduced **CAP@CS nanocapsules** into the polydopamine/sodium alginate (PDA/Alg)_m polyelectrolyte multilayer (PEM) film to make it have pH-triggered antifouling agent (CAP) release performance. When the pH value was 4 to pH 8.5, the concentration of released CAP dropped by nearly 42%, and the antibacterial effect of the multilayer film changed with the pH value as shown in Fig. 6. The design of this multilayer film provides a new idea for controlling the release behavior of antifouling agents. In addition, the process of extracting capsaicin from pepper is relatively simple and low in cost, so it is suitable for wide application.

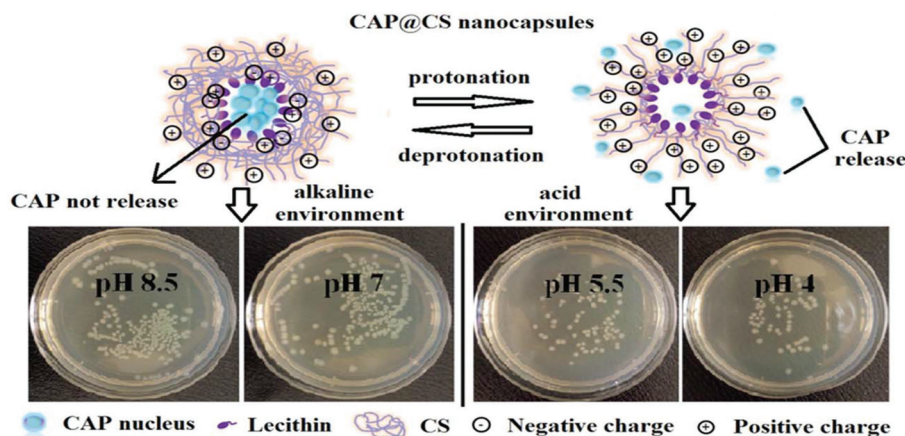


Fig. 6 The pH-triggered mechanism image of CAP@CS nanocapsules (reproduced with permission from ref. 64, Copyright 2019, Elsevier).

Tannic acid is used as an antifouling agent in antifouling coatings owing to **its inhibitory effect on microorganisms and fungi**. For example, Peres *et al.*⁶⁵ extracted tannic acid from the black wattle tree and mixed it with a soluble resin to prepare an antifouling coating. The **antifouling test shows that the coating has a good antifouling effect within 7 months**, which proves that tannic acid has good antifouling performance. Then, Xu *et al.*⁶⁶ linked the pH-responsive poly(2-diisopropyl-aminoethyl methacrylate)-*b*-poly(2-methacryloxyethyl phosphorylcholine) (PDPA-*b*-PMPC) and cationic polylysine (PLYS) to tannic acid (TA) to prepare a TA grafted binary polymer (PLYSTA-PDPA-*b*-PMPC). The result of antibacterial and antifouling functions changing with environmental pH of the coating is shown in Fig. 7. When the pH drops to 5.5, the swelling of the PDPA chain causes **zwitterionic PMPC fragments to dominate the coating surface**, thereby improving the antifouling ability of coating while reducing the antibacterial ability. When the pH rises to 7.4, the PDPA chain is broken and the coating surface will undergo rearrangement and feature switching, thus the antibacterial ability of the coating surface will be restored, but the anti-adhesion ability will decrease. Therefore, the synthetic binary polymer coating has pH-responsive switchable characteristics and self-defense capabilities.

Although the amount of antifouling agents extracted from terrestrial plants is larger than that of marine organisms, the

method of extracting antifouling agents from these organisms still cannot be used on a large scale, and these natural antifouling agents may have potential ecological toxicity. Therefore, not only the antifouling performance, but also its ecological toxicity should be explored in future research.

4.1.3. Chemically synthesized organic antifoulant. At present, copper-based antifouling agents pose certain toxicity to the marine environment, and the extraction steps of natural product antifouling agents are complicated and costly. Therefore, it is a more efficient and suitable method to prepare antifouling agents with similar or better structures than natural active substances through chemical synthesis. Currently commercial organic antifouling agents include DCOIT (4,5-dichloro-2-*n*-octyl-4-isothiazolin-3-one) and Econeal (2-(*p*-chlorophenyl)-3-cyano-4-bromo-5-trifluoromethylpyrrole).

Isothiazolinone compounds are highly efficient bactericides, which are safe and pollution-free to the environment.⁶⁷ Isothiazolinone compounds (as shown in Fig. 8a) are biosynthetic inhibitors. As a heterocyclic compound, their bactericidal mechanism mainly relies on the active part of the heterocyclic ring and DNA bases in the protein in the bacteria to form hydrogen bonds, thereby destroying the DNA structure of the cell.⁶⁸ The antifouling performance of the active ingredient DCOIT (Fig. 8b) used in the environmentally friendly Sea-Nine™ has been proven.^{69,70} However, it has been found that

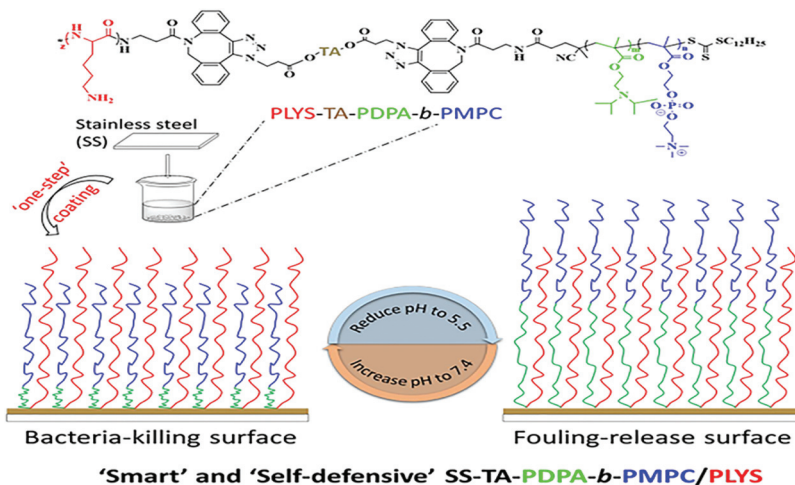


Fig. 7 Chemical structure of the tannic acid (TA)-scaffolded polymer and schematic illustration of 'one-step' deposition of 'smart' and self-defensive TA-scaffolded binary polymer brush coatings (reproduced with permission from ref. 66, Copyright 2020, American Chemical Society).

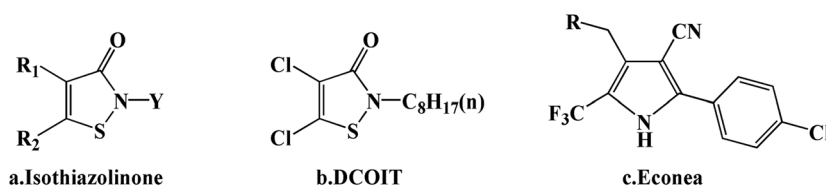


Fig. 8 The structure of isothiazolinone compounds and Econeal.

DCOIT is highly toxic to target and non-target species in practical applications. In order to reduce the toxicity of DCOIT to non-target organisms, Figueiredo *et al.*⁷¹ prepared silica nanocapsules embedding DCOIT and Ag. The encapsulated bactericide reduced its toxicity to non-target species and showed good antifouling performance.

Econea (2-(*p*-chlorophenyl)-3-cyano-4-bromo-5-trifluoromethyl-pyrrole) (Fig. 8c) is another ideal choice for the preparation of copper-free antifouling coatings because it has a broad spectrum and excellent antifouling activity against invertebrate marine life.⁷² Recently, Ferreira *et al.*⁷³ reported a non-biological agent release coating that grafted Econea biocide onto isocyanate. This kind of coating can be used for antibacterial and long-term marine anti-biological pollution. Under actual seawater conditions, the antifouling effect can be maintained for more than two years.

Although the above two antifouling agents have been proved to be new antifouling agents for marine antifouling coatings, both types of antifouling agents are monopolized by foreign countries, and their core technologies are controlled by foreign companies, so both of them are expensive. In addition, their release rate is fast and the antifouling period is short. Thus, the development of highly efficient, long-lasting, non-toxic and low-cost antifouling coatings with independent intellectual property rights has high value and significance.

5. Physical antifouling

5.1. Fouling release antifouling coating

A low surface energy antifouling coating is difficult for marine organisms to adhere to the surface of the coating film due to its low surface energy. Even if organisms adhere to the surface, they can be washed away when the ship is sailing due to their

poor adhesion, so as to achieve the effect of antifouling without the help of antifouling agents. It is generally believed that only when the surface energy of the coating is less than 25 mJ m^{-2} , or the contact angle between the coating and the liquid is greater than 98° , the coating has a good fouling release effect.⁷⁴ This type of coating is usually made of silicone and fluoropolymer.^{75–77}

5.1.1. Fouling release antifouling coating with silicone. Silicone polymer has good desorption ability, and it is easy to make marine attached organisms fall off by brushing or flowing.⁷⁸ Polydimethylsiloxane (PDMS) has the characteristics of low surface energy and low elastic modulus and gradually becomes the basis of most fouling release coatings. For example, Selim *et al.*⁷⁹ prepared polydimethylsiloxane/ZnO (PDMS/ZnO) nanorod (NR) composites by an *in situ* method. The superhydrophobic and self-cleaning behavior of the fabricated PDMS/ZnO (0.5 wt%) NR composites that prevent fouling adhesion is shown in Fig. 9a. It is found that after 0.5 wt% nanorods are distributed on the surface, the non-wettability of the coating is enhanced, the surface roughness becomes larger, the surface free energy is the smallest (11.25 mN m^{-1}), the WCA is the largest (158°) (Fig. 9b and c), and the coating achieves the best antifouling performance among the 7 samples.

5.1.2. Fouling release antifouling coating with organic fluorine. In fluoropolymers, the material has high chemical stability and oil or water repellency owing to the high electronegativity of fluorine atoms, low degree of polarization, and high C–F bond energy.⁸⁰ Recently, Xie *et al.*⁸¹ obtained low surface energy fluorine, silicon modified marine antifouling coating which used the telomer of dodecafluoroheptyl methacrylate (DFMA), triclosan acrylate (TA) and 3-mercaptopropyltrimethoxysilane (KH590) to graft bisilanol-terminated PDMS. The synthetic route is shown in Fig. 10. In fact, due to the

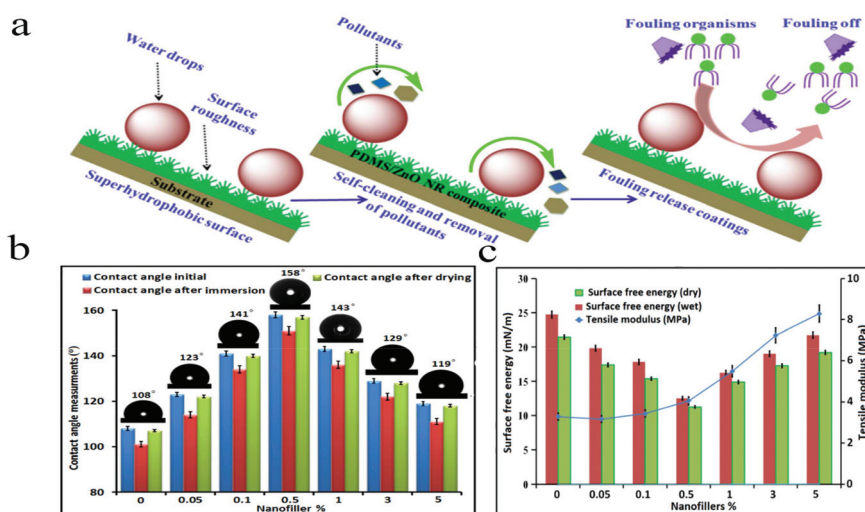


Fig. 9 (a) The superhydrophobic and self-cleaning behavior of the fabricated PDMS/ZnO (0.5 wt%) NR composites that prevent fouling adhesion. (b) Represents WCA values of the silicone/ZnO NRs composite films before and after water submersion for a week and for the dried film; (c) surface free energy evaluations of silicon/ZnO NR composites before and after submersion (for a week) in demineralized H₂O (reproduced with permission from ref. 79, Copyright 2018, Elsevier).

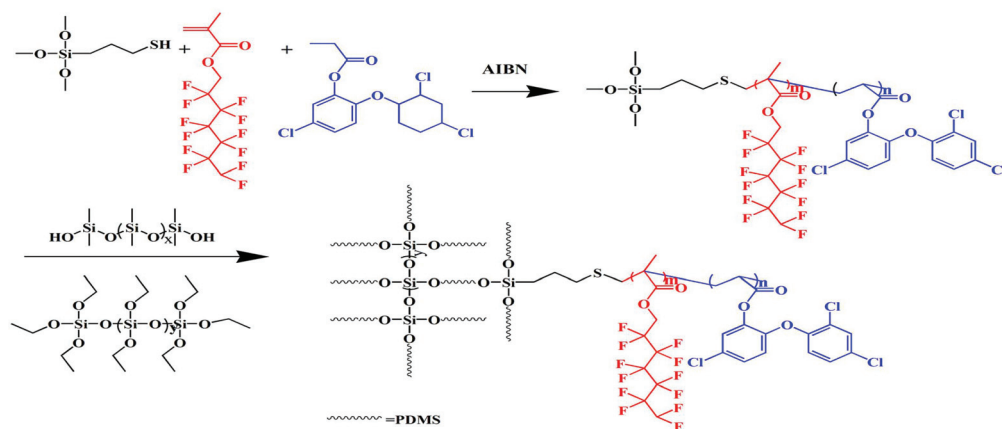


Fig. 10 Synthesis of PFTA and PDMS-FTx. (Reproduced with permission from ref. 81, Copyright 2019, John Wiley and Sons.)

migration of DFMA, the antifouling groups are enriched on the coating surface, so the surface has antifouling performance. In addition, the surface of each coating has low surface

energy and the WCA is above 105°. Therefore, the coating has good antifouling release performance. Table 2 shows the composition, the weight ratio and names of samples. The results of marine field tests are shown in Fig. 11. After being immersed in seawater for 3 months, PDMS-FT4, PDMS-FT8 and PDMS-FT12 showed high antifouling efficiency, with only a few organisms attached. Most of the attached biofouling could be washed away after applying a water jet to the surface of the coating, indicating that the materials have good fouling release performance. The coatings were seriously polluted as the dirt pressure intensified within 6 months. After applying water jet, the surfaces of PDMS-FT8 and PDMS-FT12 were almost clean. This shows that the combination of antifouling and fouling release performance of the graft copolymer PDMS coating promotes its long-term antifouling performance.

The low surface energy coatings do not need to use antifouling agents, and the surface is generally smooth, which is of

Table 2 Composition of PDMS-based coatings

Sample	Bis-silanol [g]	PFTA/PTA ^a	ES-40 ^b [g]	DFMA content [%]	TA content [%]
PDMS	10.00	00	0.25	0	0
PDMS-T4	10.00	0.40	0.25	0	3.7
PDMS-FT4	10.00	0.40	0.25	2.2	1.6
PDMS-FT8	10.00	0.40	0.25	4.2	3.0
PDMS-FT12	10.00	0.40	0.25	6.1	4.4

^a PTA (telomer of TA and KH590, molar ratio is 9.3 : 1) was added into PDMS-T4 and PFTA (telomer of DFMA, TA and KH590, molar ratio is 4.1 : 4.3 : 1) was added into PDMS-FTx. ^b ES-40: poly(diethoxy siloxane).

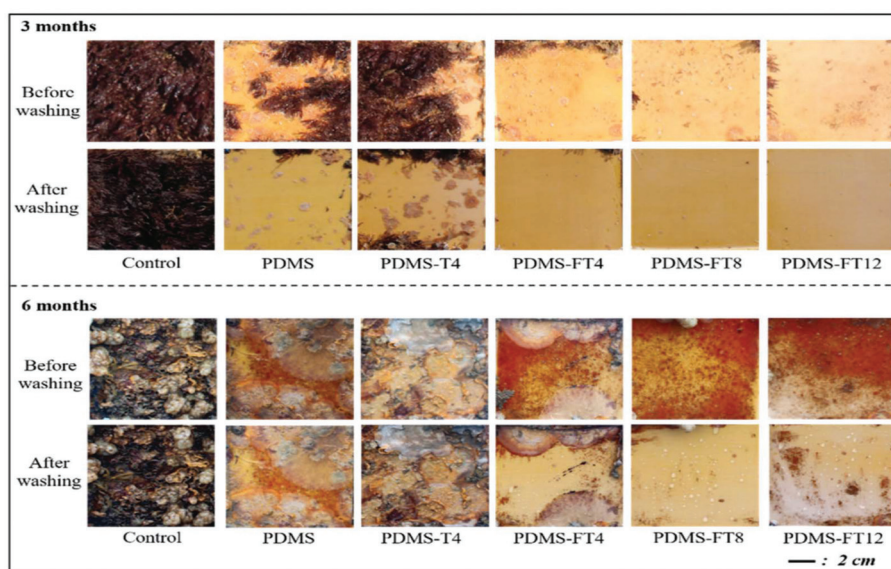


Fig. 11 Pictures of PDMS-based coatings after immersion in Xiamen Bay, China before and after a water jet treatment (pressure: 2 MPa) (reproduced with permission from ref. 81, Copyright 2019, John Wiley and Sons).

great significance for reducing the sailing resistance of ships and fuel consumption. The obvious disadvantage of low surface energy coatings is poor adhesion to the hull surface. In practical applications, an intermediate coating must be used to enhance the adhesion of the coating, which complicates the coating process and increases the cost of use. This is also the problem that researchers should solve in future research.

5.2. Fouling resistance antifouling coating

Fouling resistance materials refer to materials that can inhibit and prevent the adhesion and growth of marine organisms. They are usually hydrophilic polymers as a result of the low interface energy between the polymers and water, and a hydration layer usually can be formed on the surface. When organisms approach, they have to break through the hydration layer to bond with the surface of the substrate, which leads to the need for more energy, thus reducing the possibility of adhesion. These types of coatings mainly include polyethylene glycol, hydrogels and zwitterionic polymers, and have good resistance to proteins, marine bacteria, green algae spores and barnacle larvae.^{82–84}

5.2.1. Polyethylene glycol polymer. Polyethylene glycol (PEG) has a large removal volume, strong mobility of hydration chains and steric hindrance, which can effectively reduce cell adhesion and growth and protein adsorption, and the antifouling effect is better when the degree of polymerization is higher.^{85,86} For example, Kim *et al.*⁸⁷ used an anthracene (ANT)-attached polyethylene glycol (PEG) surface modification agent (SMA) to prepare a polyethylene terephthalate (PET) base film, and PEG chains are covered on the surface of the material in a brush-like and arch-like form to produce a hydration layer (Fig. 12). The results show that the modified coating has higher ion permeability and resistance to organic pollution.

5.2.2. Hydrogel polymer. A hydrogel (cross-linked by hydrophilic polymer chains, such as polyethylene glycol or polyhydroxyethyl methacrylate)^{88–91} and silicone oil^{92,93} can form a hydration layer or lubricating layer on the surface of the

coating to reduce fouling. In addition, the swelled hydrogel has the characteristics of softness and high elasticity, and most marine organisms prefer to attach to hard surfaces. Therefore, the hydrogel coating has good antifouling properties. For example, Murosaki *et al.*^{94,95} studied the antifouling behavior of a series of synthetic hydrogels on barnacles in long-term marine environment experiments and proved the effective antifouling properties of the hydrogels. However, the strong mechanical and physicochemical properties of the hydrogel and how to coat the surface of the substrate are important for practical applications. Unfortunately, the current research on the application of hydrogel marine antifouling is very limited. Therefore, Zhu *et al.*⁹⁶ reported the simple preparation of a hydrogen-bonded supramolecular polyvinyl alcohol (PVA)–glycerin gel and proposed the use of a poly(allylamine salt) hydrogel (PAH) and α -ethyl cyanoacrylate (ECA) as an adhesive between the hydrogel and the substrate. The final coating has excellent thermoplasticity and mechanical properties, and the PVA–glycerin hydrogel has a strong settling inhibitory effect on wisteria.

5.2.3. Zwitterionic polymer. Zwitterionic polymers, such as sulfobetaine, carboxybetaine and phosphocholine, can contain both cations and anions in the structure.^{85,97} Different from the hydration layer formed by hydrogen bonding of non-ionic hydrophilic polymers (such as PEG), a zwitterionic polymer can induce hydration through electrostatic interaction, which allows it to bind tightly to water and therefore obtain a better antifouling effect. Dai *et al.*⁹⁸ prepared a new type of hydrolysis-inducing zwitterionic monomer-tertiary carboxybetaine triisopropylsilyl ethyl acrylate (TCBSA) and copolymerized it with methyl methacrylate (MMA). This copolymer rapidly self-generates the zwitterionic surface due to hydrolysis, and the polymer chain can be dissolved in seawater to form a self-renewing dynamic surface (Fig. 13). Tests show that the coating has good resistance to seawater protein in the marine environment and can effectively inhibit the adhesion of marine bacteria (*Pseudomonas* sp.) and diatoms (*Navicula incerta*).

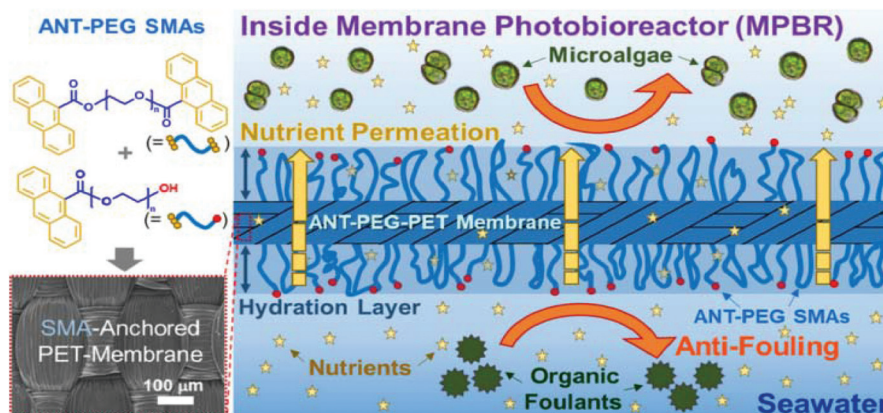


Fig. 12 Illustration of a photobioreactor equipped with the PET membranes surface-modified by ANT–PEG SMAs (reproduced with permission from ref. 87, Copyright 2020, American Chemical Society).

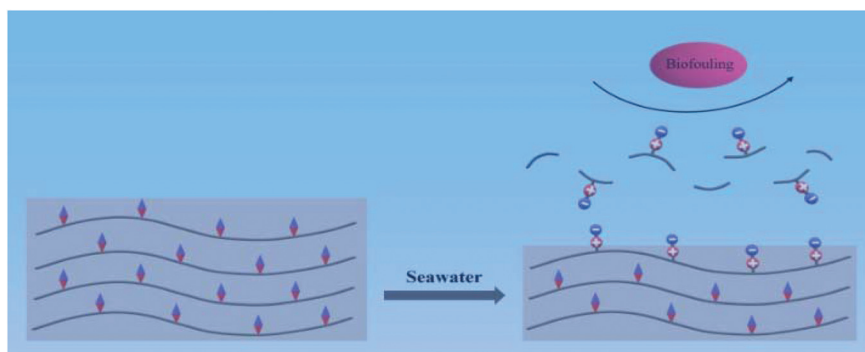


Fig. 13 The antifouling mechanism of the coating (reproduced with permission from ref. 98, Copyright 2019, American Chemical Society).

Although fouling-resistant materials show resistance to a variety of marine bacteria and diatoms in the laboratory simulated marine antifouling test, their antifouling broad spectrum is poor, and it has no antifouling effect on some large fouling organisms as a result of the complexity of the marine environment and the diversity of marine fouling organisms. In addition, when this type of coating is applied to the marine environment, the antifouling property will fail once the surface of the coating is covered by marine silt. Therefore, the use of such materials alone has great limitations; combining multiple antifouling methods is an inevitable trend in the development of marine antifouling coating.

5.3. Bionic coating

5.3.1. Micro-nano structured surface. Nature is the best mentor of mankind, and the surfaces of many marine creatures (sharks, dolphins, *etc.*) are hardly parasitized by other marine foulings. Studies believe that this is related to the surface microstructure. The microstructured surface makes the fouling organisms have fewer attachment points on it, and the adhesion is not firm.^{99,100} Early researchers used laser ablation, photolithography and mold casting to etch micro-nano structures on silicone, polycarbonate, polyvinyl chloride, polyimide, *etc.* and studied their antifouling effect. For example, Brennan *et al.*^{101,102} prepared a microstructured surface that mimics shark skin and studied the adhesion of *Ulva* spores on the surface. It is found that when the microstructure gap on the surface is smaller than the spore length of *Ulva* Spores, the adhesion rate of spores on the surface is lower than that on a smooth surface. Then, Gatenholm *et al.*¹⁰³ found that the adhesion rate of barnacles on the surface of unmodified polyvinyl chloride (PVC) is very high, but PVC with a micro-nano structure can significantly reduce the adhesion of barnacles. In addition, Genzer *et al.*¹⁰⁴ believed that a surface with a single microstructure is difficult to use as a broad-spectrum antifouling material as the size of the fouling organisms varies greatly.

Modern studies have shown that the microstructured surface may only be one of the factors affecting the antifouling ability of sharks and dolphins, and the self-shedding of the surfaces and the secretion of mucus substances also promote

its antifouling.^{19,105–109} For example, studies have found that live fish scales have long-lasting antifouling properties due to their micro-nano structure and mucus regeneration. But when the fish dies, the surface of the scale gradually loses its antifouling properties owing to the biodegradation of the mucus. In order to maintain the long-lasting anti-fouling performance of underwater coatings, the best way is to give it a **self-renewal ability similar to living organisms**. Wang *et al.*¹¹⁰ were inspired to **embed seawater-responsive polymer-modified silica nanoparticles (PTM-SiO₂) in a self-polishing polymer (SP) matrix**. As shown in Fig. 14, after immersing the coatings in artificial seawater (ASW) for 2 days, the relatively tight SP/25PTM-SiO₂ (the content of PTM-SiO₂ is 25%) coating becomes porous, creating a unique micro-nano structure. The surface of the coating realizes self-renewal through the hydrolysis of the surface self-polishing polymer SP, then PTM-SiO₂ is exposed on the surface of the coating, and a micro-nano structure is established; thereby the long-term antifouling performance is achieved.

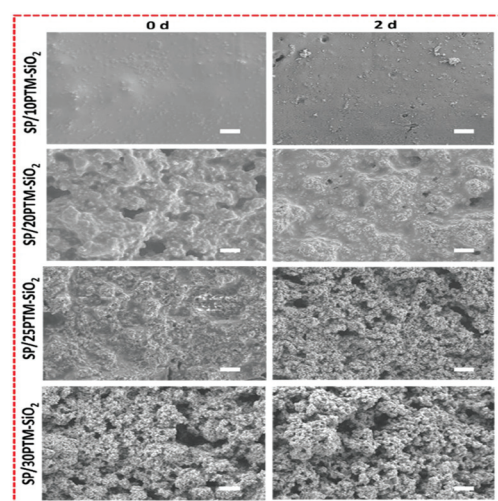


Fig. 14 FE-SEM images of the surface of SP/xPTM-SiO₂ (x = the nano-particle content) coatings before and after submersion in ASW for 2 days. The scale bar is 1 μm. (Reproduced with permission from ref. 110, Copyright 2018, American Chemical Society.)

5.3.2. Slippery liquid-infused porous surfaces. Inspired by the method of trapping insects by *Nepenthes*, the researchers injected lubricating fluids such as fluoride or silicone oil into the material with a micro-nano surface structure to make the coating have a “super-slip surface”. The obtained coating will be difficult for fouling to adhere. This is called Slippery Liquid-Infused Porous Surfaces (SLIPS).^{111–117} For example, Aizenberg *et al.*^{114,118} reported for the first time that by injecting a perfluorinated lubricant liquid into the surface of a porous polytetrafluoroethylene film, the prepared SLIPS has a low contact angle hysteresis, which can effectively inhibit the sedimentation of common bacteria such as *Staphylococcus aureus* and *Escherichia coli* on the surface.

However, in practical applications, the continuous lubricating layer of traditional SLIPS relies on capillary force, which is prone to damage after severe mechanical damage to the porous substrate.¹¹⁹ It is inferred that a stable smooth surface can be designed from two aspects: **the self-healing lubricant layer and the self-healing substrate.** Zhang *et al.*¹²⁰ used α,ω -aminopropyl polydimethylsiloxane (APT-PDMS) and isophorone diisocyanate (IPDI) to synthesize polydimethylsiloxane (PDMS-PUa), and then injected silicone oil into PDMS-PUa to design a smooth organic gel layer (OG). The self-healing properties of the substrate are attributed to the breaking and reforming of the hydrogen bond of the urea group between PDMS-PUa, and the self-generation of the lubricating oil layer is the result of penetration of silicone oil into the damaged area through the PDMS-PUa substrate. The experimental results show that the prepared OG has good lubricating performance and excellent performance in inhibiting bacterial sedimentation (Fig. 15c). The fractured coating can realize self-healing after being in contact for 48 hours at room temperature, and the healed specimen can withstand external forces (the mass of weight = 500 g) without any deformation (Fig. 16).

Bionic antifouling coatings do not contain antifouling agents and will not release compounds into seawater; therefore they are an environmentally friendly antifouling coating. However, the structure of the surface microstructure is more complex and it is difficult to repair after damage; thus the antifouling effect is poor in the actual marine environment. With

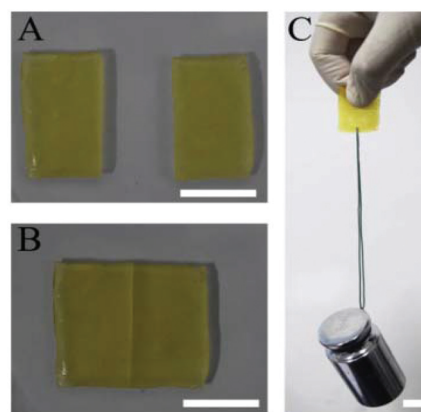


Fig. 16 The self-healing process of OG at room temperature (scale bar = 1 cm). (A) damaged sample, (B) healed sample, (C) healed sample can withstand the external force (the mass of weight = 500 g). (Reproduced with permission from ref. 120, Copyright 2019, Elsevier).

the development of repair technology, bionic antifouling coating will be still a promising development direction.

6. Novel antifouling materials

6.1. Novel composite coating

6.1.1. Nanocomposite antifouling coating. Nanocomposite coating is a new type of antifouling coating with various advantages by incorporating antifouling agents such as nanocuprous oxide (Cu_2O NPs), nanosilver (Ag NPs) and nanosilica (SiO_2 NPs) into low surface energy, bionic, natural product coatings.^{121–124} For example, Selim *et al.*¹²⁵ prepared a superhydrophobic PDMS-Ag@ SiO_2 core-shell nanocomposite antifouling coating. The formation process of the coating is shown in Fig. 17(a). A solution casting method is used to embed the Ag@ SiO_2 core-shell nanofiller on the surface of the PDMS material and form a coating after curing into a film. The water contact angle (WCA) of the coating is 156° , and the surface free energy is 11.15 mJ m^{-2} (Fig. 17(b)). The composite material has excellent superhydrophobicity, a self-cleaning effect and low surface energy, and has certain inhibitory properties to different bacterial strains, yeasts and fungi.

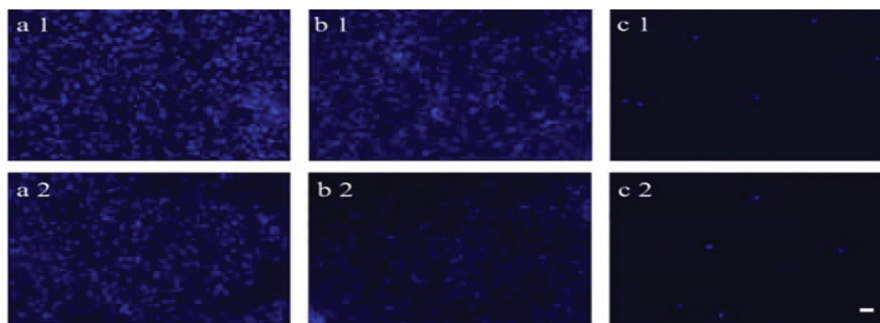


Fig. 15 The fluorescence microscopy images of samples after immersion in static sea water 1 and dynamic sea water 2 inoculated with *Pseudoalteromonas* sp. for 7 days; (a) bare glass (BG), (b) PDMS-PUa, (c) OG (reproduced with permission from ref. 120, Copyright 2019, Elsevier).

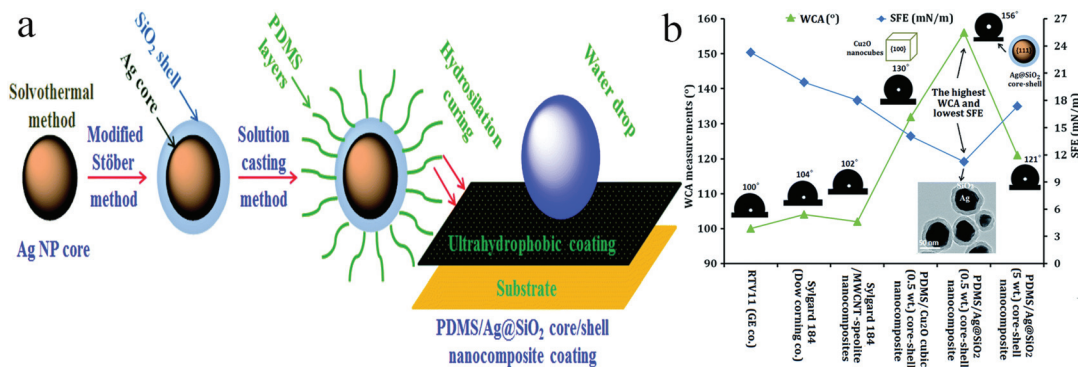


Fig. 17 (a) Schematic diagram of the Ag@SiO₂ core-shell nanosphere structure, and (b) comparable study of the FR and self-cleaning performance of PDMS/spherical Ag@SiO₂ core-shell nanocomposites and other commercially developed and sounded FR surfaces (reproduced with permission from ref. 125, Copyright 2018, Royal Society of Chemistry).

A nanocomposite (NC) hydrogel is composed of nanoparticles and grafted polymer chains. Its application in antifouling has aroused great interest.^{126–128} However, studies have shown that the NC hydrogel is not sprayed directly on the solid surface for antifouling due to its low mechanical strength and poor adhesion to solids. Adding NC hydrogels to commercially available silicone elastomers to develop new antifouling coatings is a new idea for expanding their applications.¹²⁹ Thus, Tian *et al.*¹³⁰ developed a series of good performance NC hydrogels chelated with sulfhydryl/acetylthioester groups and nanosilver (AgNPs) and combined them with organic silicon substrates to develop a new type of antifouling nanocomposite coating. Using AgNPs as a cross-linking agent, the hydrophilic polymer can be well dispersed in the silicone resin matrix and compatibility of silicone and hydrogels can be enhanced. The results of the tests described that the prepared coating has better antifouling performance and its adhesion to the substrate is getting improved.

In previous studies, many researchers have studied the antibacterial effects of nanoparticles by doping single nanoparticles, such as AgNPs, TiO₂ and ZnO, but there are few reports on the improvement of material properties by doping multiple nanoparticles in the coating. Thus, Verma *et al.*¹³¹ prepared SiO₂, TiO₂ and SiO₂-TiO₂ core-shell nanoparticles by the sol-gel method respectively, and the coating formulations were solvent-based polyurethane and water-based polyacrylic acid. The results show that the polyurethane coating with 4 wt% SiO₂-TiO₂ core-shell-shell nanoparticles has the best antibacterial effect, and has good adhesion and impact resistance. This research provides a new idea for the development of nanocomposites.

In recent years, nano-composite antifouling coatings have attracted more and more attention, which have the advantages of multiple antifouling technologies. They have the characteristics of enhancing the stability of the nanoparticles in the coating, improving mechanical properties, water resistance, abrasion resistance, and antifouling performance of the coating, and prolonging the antifouling cycle. The nano-composite antifouling coatings provide a new way for future research of new marine antifouling coatings.

6.1.2. Photocatalytic antifouling coating. The free radicals generated by the photocatalytic material under visible light irradiation have a good effect on the degradation of pollutants and the sterilization of bacteria. Therefore, photocatalytic antifouling technology can be used for marine antifouling.¹³²

For example, vanadium oxalate is a chemical raw material, and it is widely used in the preparation of nitric acid oxidation catalysts and metal complexes.^{133,134} Zhang *et al.*¹³⁵ proposed a composite photocatalyst of indium vanadate/silver vanadate (InVO₄/AgVO₃). Compared with pure InVO₄ and pure AgVO₃ crystals, its photocatalytic antifouling efficiency is significant. Nevertheless, the InVO₄/AgVO₃ composites are expensive and not suitable for large-scale use. Influenced by this, Zhang changed the employed silver to bismuth. Bismuth vanadate (BiVO₄) is an ideal photocatalytic material that decomposes water and degrades organic pollutants under visible light irradiation. Thus, Zhang *et al.*¹³⁶ made an in-depth study and synthesized a bismuth vanadate/indium vanadate (BiVO₄/InVO₄) nanocomposite by the hydrothermal method. The action mechanism of the material is shown in Fig. 18. When the sun shines on the material, it will excite carriers such as electrons and holes. The electrons in the conduction band (CB) of InVO₄ will enter into the CB of BiVO₄ and then react with O₂ to generate [•]O₂[−]. Similarly, the holes in the valence band (VB) of BiVO₄ will migrate to the VB of InVO₄ and then react with the OH[−] to generate [•]OH radicals. The resulting [•]O₂[−] and OH will damage the cell wall and cause the release of DNA molecules, and then the cell will completely be decomposed. Thus the material antifouling performance is improved. The test results show that the degradation efficiency of the material to model pollutants (rhodamine B, RhB) reaches 99.775% within 280 min (Fig. 19a), and the inhibition rate against common bacteria such as *E. coli* and *Staphylococcus aureus* exceeds 99.5% (Fig. 19b). It shows that the synthesized material has excellent photocatalytic antifouling performance.

Although photocatalytic technology has made great progress through the efforts of researchers, it is still in the stage of theoretical exploration and experimentation as a whole and has not yet reached the industrial scale; therefore its practical

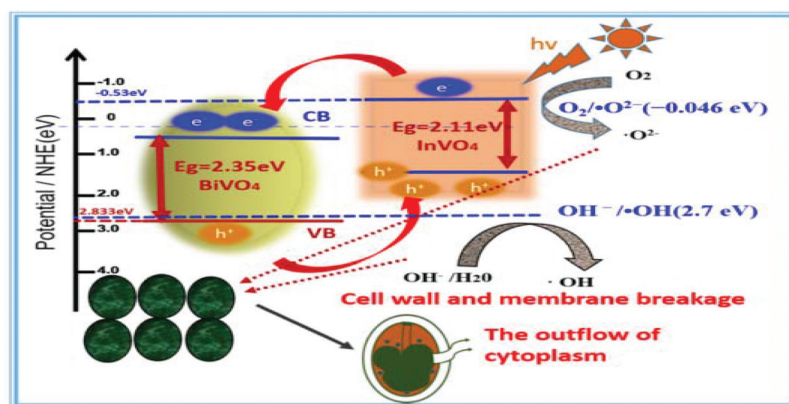


Fig. 18 Schematic diagram of the proposed mechanism for the photo-generated charge carrier transfer in the 0.2InVO₄–0.8BiVO₄ composition under visible light irradiation (reproduced with permission from ref. 136, Copyright 2019, Elsevier).

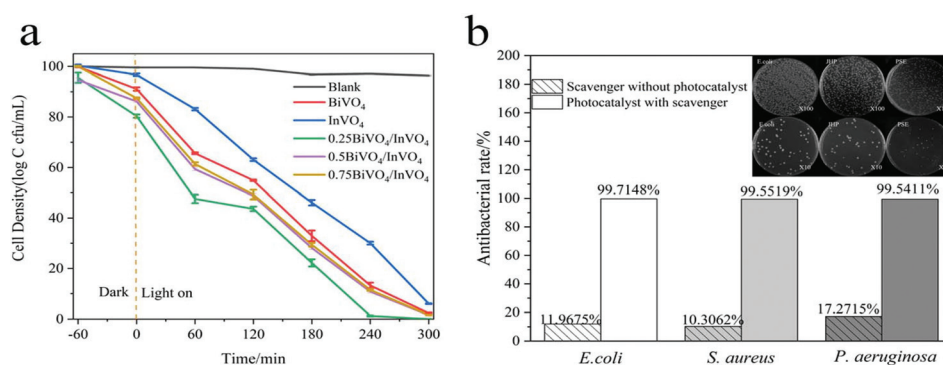


Fig. 19 (a) Degradation efficiencies of BiVO₄/InVO₄, and (b) photo-catalytic disinfection rate histogram of *E. coli*, *S. aureus*, and *P. aeruginosa* under visible light (reproduced with permission from ref. 136, Copyright 2019, Elsevier).

application has been greatly restricted. The main problems include two aspects, one is the low efficiency of carrier transfer, and the other is the low utilization of visible light. Therefore, it is necessary to continuously improve the existing problems and enhance the activity and stability of photocatalysis in future research, which is essential for the development of photocatalytic antifouling technology.

6.1.3 Amphiphilic antifouling coating. One of the most popular antifouling research directions is amphiphilic marine antifouling coatings. The presence of hydrophilic groups in amphiphilic coatings is conducive to hydration, thereby inhibiting protein adsorption. The hydrophobic group can reduce the surface energy of the material, then reduce the polar bond and hydrogen bond interaction with the bioadhesive; thereby biofouling can be easily removed.

In the early days, research studies on synthetic raw materials for amphiphilic antifouling coatings composed of hydrophilic and hydrophobic materials mainly focused on the covalent cross-linking of polyethylene glycol (PEG) bases and fluoropolymers.^{137–139} However, PEG is easily oxidized, which limits its practical application in the marine environment.¹⁴⁰ A zwitterionic material is a promising hydrophilic antifouling

material, different from PEG, the zwitterionic polymer can induce hydration through electrostatic interaction, which makes it tightly bound to water, so it has a better antifouling effect. It is also considered to be a suitable substitute for PEG in amphiphilic coatings.¹⁴¹ For example, Ruiz-Sanchez *et al.*¹⁴² prepared a cross-linked amphiphilic membrane composed of perfluoropolyether and zwitterionic monomer methacrylic acid sulfobetaine (Fig. 20). The result shows that this type of polymer film has good antifouling surface properties and can effectively play the dual role of fouling resistance and release.

Although amphiphilic polymers have proven that the combination of hydrophobic and hydrophilic side chains can improve the antifouling performance of marine antifouling polymer coatings, studies on the side chains of amphiphilic polymers that contain both hydrophobic and hydrophilic groups have not yet been performed. Thus, Xie *et al.*¹⁴³ prepared an environmentally friendly marine antifouling coating HO-FPTU through the polymerization reaction with hexamethylene diisocyanate (HDI), pentaerythritol tetrakis (3-mercaptopropionate) (PETMP) and 2,3,5,6-tetrafluoro-1,4-benzene-dimethanol (HOCH₂-FB-Al). The antifouling mechanism of this coating is that the side chains containing hydrophobic

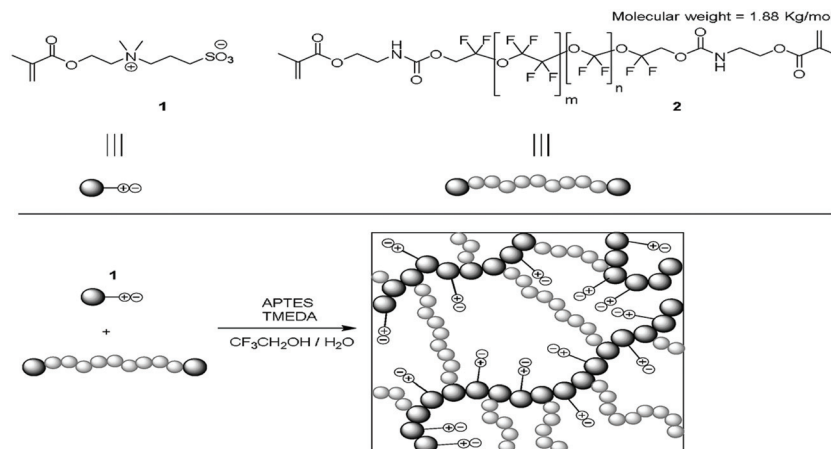


Fig. 20 Structures of sulfobetaine methacrylate and α,ω -bis-methacrylate PFPE (reproduced with permission from ref. 142, Copyright 2019, Elsevier).

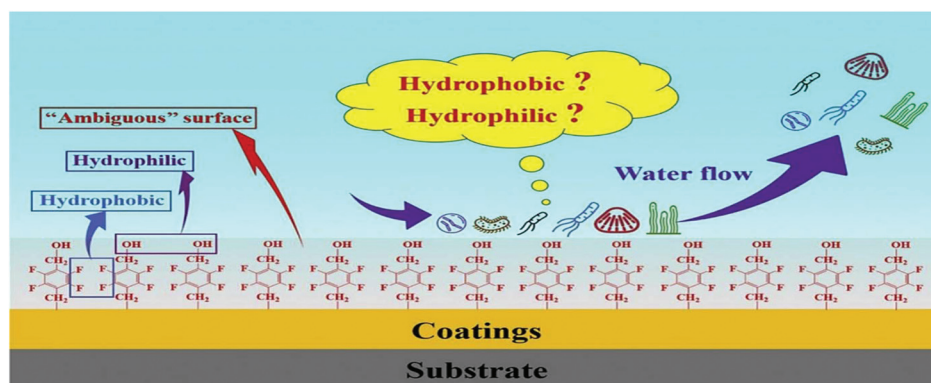


Fig. 21 Schematic antifouling mechanism of HO-FPTU coatings (reproduced with permission from ref. 143, Copyright 2020, Elsevier).

groups and hydrophilic groups are distributed on the surface of the HO-FPTU polymer coating to form a “fuzzy” surface, and marine life may be confused by this “fuzzy” surface and can be easily removed during the attachment process (Fig. 21). The test results show that the coating has **good protein resistance and algae resistance and it still has good antifouling performance even if it lasts for 6 months under static conditions in the marine environment.**

Among the many environmentally friendly marine antifouling coatings, amphiphilic coatings are becoming a very popular research trend. And the amphiphilic dynamic surface has good research and application prospects for inhibiting the attachment of fouling organisms and enhancing their release ability.

7. Conclusion and prospect

The marine fouling problem encountered by humans in the process of exploring the ocean causes huge economic losses every year, and painting with antifouling coatings is currently

the most effective method to prevent marine organisms from fouling. As people's awareness of environmental protection has gradually increased, marine antifouling coatings are developing in the direction of strong antifouling capabilities, simple processes, safety and environmental protection. Therefore, the future research trends of marine antifouling technology will mainly focus on the following aspects. Firstly, marine antifouling coatings should be developed in the direction of environmental protection and non-toxicity. Therefore, the potential ecotoxicity of antifouling coatings must be studied in depth, and harmful antifouling coatings will be eliminated. Secondly, the application of nanotechnology has enabled the development of marine antifouling technology to achieve a qualitative leap. Adding nano-scale materials to coatings or making the key components of the coating nano-scale will enhance the related properties of the coating, such as mechanics and antifouling performance. In addition, functional groups with different antifouling effects are added to the coating by block copolymerization, which can effectively improve the antifouling performance of coatings. On the whole, coatings that rely on a single approach cannot achieve

the desired antifouling effect in the practical applications, and the focus of future studies will be integrating multiple methods.

Conflicts of interest

The authors declare no conflict of interest.

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